First correction to the critical mean degree in extreme long-range percolation

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Abstract

Consider independent undirected bond percolation on the integer lattice, with connection probability $C|x-y|^{-1-\sigma}$, where $0<\sigma<1$. We determine the first correction to the critical expected degree as σ tends to zero. Writing $p_c = 2\zeta(1+\sigma)C_c(\sigma)$, we prove

$$p_c = 1 + \frac{\zeta(2+2\sigma)}{2\zeta(1+\sigma)^2} + O(\sigma^3) = 1 + \frac{\pi^2}{12}\sigma^2 + O(\sigma^3).$$

The error is uniform for all sufficiently small positive real σ, with dimension fixed and no additional spread parameter. The lower bound follows from an exact nonbacktracking path estimate. For the upper bound, elementary Fourier estimates verify the general small-triangle hypotheses for the lace expansion. We then retain a deleted-bond restriction in the first lace coefficient and distinguish a closed diagram of order σ^2 from its spatial supremum of order σ. The positive remainder after the first odd truncation converts a uniform subcritical coefficient bound into the threshold bound. This avoids a critical limit of lace coefficients and an estimate of the entire higher-order lace tail in the additional correction calculation; the imported infrared theorem itself still relies on lace-expansion estimates. We extract a two-scale kernel criterion and apply it uniformly to arbitrary nonnegative summable additions and compact mixtures of nearby tail exponents. For an added nearest-neighbor weight λ, the coefficient becomes $\pi^2/12 + \lambda + \lambda^2/2$; for exponent mixtures it is multiplied by the square of the mean exponent ratio.

Keywords: long-range percolation; critical probability; lace expansion; nonbacktracking paths; small-exponent asymptotics.

1 Model and result

For $0<\sigma<1$ and $0\le C\le 1$, declare each unordered pair $\{x,y\}$ of distinct integers open independently with probability $C|x-y|^{-1-\sigma}$. Let $C_c(\sigma)$ be the infimum of the parameters for which the origin belongs to an infinite open cluster with positive probability. At $C=1$ all nearest-neighbor bonds are open, so this infimum is taken over a nonempty set. Put

$$Z_\sigma = \zeta(1+\sigma), \qquad D_\sigma(0)=0, \qquad D_\sigma(x) = \frac{|x|^{-1-\sigma}}{2Z_\sigma} \quad (x\neq 0). \tag{1}$$

Thus D_σ is an even probability distribution. The normalized parameter $p = 2Z_\sigma C$ is the expected degree, and the bond probabilities become $pD_\sigma(y-x)$. Its admissible interval is $[0, \|D_\sigma\|_\infty^{-1}] = [0, 2Z_\sigma]$. We write $p_c(\sigma) = 2Z_\sigma C_c(\sigma)$.

Theorem 1.1. *There are constants $K<\infty$ and $\sigma_0>0$ such that, for every real $0<\sigma<\sigma_0$, the model defined above satisfies*

$$|p_c(\sigma) - 1 - S_2(\sigma)| \le K\sigma^3, \qquad S_2(\sigma) = \frac{\zeta(2+2\sigma)}{2\zeta(1+\sigma)^2}. \tag{2}$$

In particular,

$$\lim_{\sigma\downarrow 0}\frac{2\zeta(1+\sigma)C_c(\sigma)-1}{\sigma^2}=\frac{\pi^2}{12}. \tag{3}$$

In the original bond-amplitude parameter the same result reads

$$C_c=\frac{1}{2Z_\sigma}+\frac{\zeta(2+2\sigma)}{4Z_\sigma^3}+O(\sigma^4),\qquad C_c-\frac{1}{2Z_\sigma}=\frac{\pi^2}{24}\sigma^3+O(\sigma^4). \tag{4}$$

Indeed, divide (2) by $2Z_\sigma$ and use $Z_\sigma^{-1}=\sigma+O(\sigma^2)$. This is an asymptotic improvement over the branching bound, not an evaluated error guarantee at a specified positive exponent.

The leading limit $p_c\to 1$ is raised for this undirected model in [1, Section 7]. That paper develops explicit upper bounds using monotone paths, which also apply to the oriented problem. The correction in (2) concerns the undirected model and its exact linear Bernoulli bond probabilities.

The term $D^{*2}(0)$ is familiar from critical-point expansions. In a spread-out regime above the upper critical dimension, van der Hofstad and Sakai [4, Theorem 1.1, (1.14)] obtain

$$p_c=1+D^{*2}(0)+\frac{1}{2}\sum_{n\geq 3}(n+1)D^{*n}(0)+O(\beta^2),\qquad \beta=L^{-d}.$$

That theorem fixes the profile and takes the spread parameter L large; its percolation statement assumes $d>6$. It does not directly cover (1) with dimension one and $\sigma\downarrow 0$. Furthermore, the general lace-expansion parameter here must dominate $\|D_\sigma\|_\infty=\Theta(\sigma)$, so an error of its square would be too large to determine (3).

We use the kernel-general infrared theorem of Heydenreich, van der Hofstad and Sakai [3], and the first-coefficient events of Borgs, Chayes, van der Hofstad, Slade and Spencer [2]. The additional estimates below keep the dependence on σ explicit. The two restrictions that matter are the original first attachment being nonzero, and a terminal vertex lying outside a deleted-bond cluster. They leave a single main term in which deletion of the first bond removes one of two apparent two-step contributions. Only a one-sided bound on this coefficient is required.

Throughout, $f*g$ denotes convolution on $\mathbb{Z}$, f^{*m} a convolution power, and $\delta(x)=\mathbf{1}_{\{x=0\}}$. Fourier transforms are $\widehat{f}(k)=\sum_x f(x)e^{ikx}$, with normalized measure $\mathrm{d}k/(2\pi)$ on $[-\pi,\pi]$. All constants denoted by c or K are finite and may increase from line to line. Unless additional indices are shown, they are independent of σ and of $p\in[1,p_c(\sigma))$ on one sufficiently small fixed interval $0<\sigma<\sigma_0$. No claim of uniformity in growing convolution orders is made.

2 Uniform estimates for the power-law kernel

Write $s=\sigma$, $Z=Z_s$, $D=D_s$, and $d(k)=\widehat{D}_s(k)$ within proofs. Integral comparison gives

$$s^{-1}\leq Z\leq 1+s^{-1}. \tag{5}$$

In particular $\|D\|_\infty\leq s/2$. The following estimates also control the very small Fourier frequencies, where pointwise convergence of d to zero would not suffice.

Lemma 2.1. *Let $L_0=\pi+1+\log 2$. For $0<s\leq 1/2$ and $0<|k|\leq\pi$,*

$$|d(k)|\leq\min\{1,s[L_0+\log_+(1/|k|)]\}, \tag{6}$$

$$1-d(k)\geq\frac{1}{3}\left(\frac{|k|}{2\pi}\right)^s. \tag{7}$$

For fixed integers $r\geq 1$, $q\geq 1$, there is a constant $K_{r,q}$ such that

$$J_{r,q}(s):=\int_{-\pi}^{\pi}\frac{|d(k)|^r}{[1-d(k)]^q}\frac{\mathrm{d}k}{2\pi}\leq K_{r,q}s^r\quad(0<s\leq\min\{1/2,1/(2q)\}). \tag{8}$$

The corresponding estimate with $q=0$ holds for $0<s\leq 1/2$.

Proof. By symmetry it is enough to consider $0 < k \le \pi$. Partial sums of e^{ink} over an interval have modulus at most $1/\sin(k/2)$. Abel summation against the decreasing sequence n^{-1-s} therefore gives

$$\left|\sum_{n=N}^{\infty} n^{-1-s} e^{ink}\right| \le \frac{N^{-1-s}}{\sin(k/2)} \le \frac{\pi}{k} N^{-1-s} \qquad (N \ge 1). \tag{9}$$

For $k \le 1$, split the series defining $Zd(k)$ at $N = \lceil 1/k \rceil$. The head is at most $1 + \log N \le 1 + \log 2 + \log(1/k)$ in absolute value, and (9) bounds the tail by π. For $k \ge 1$, use (9) with $N = 1$. Division by $Z \ge 1/s$ gives the second bound in (6); the bound by one follows because d is a characteristic function.

For the spectral gap, take $N = \lceil \pi/k \rceil$, so that $\pi/k \le N \le 2\pi/k$. Positivity of $1 - \cos(nk)$ and the tail integral give

$$\begin{aligned} Z[1 - d(k)] &\ge \sum_{n=N}^{\infty} n^{-1-s}[1 - \cos(nk)] \\ &\ge \frac{N^{-s}}{s} - \frac{\pi}{k} N^{-1-s} \ge N^{-s}(s^{-1} - 1). \end{aligned}$$

Since $(1 - s)/(sZ) \ge (1 - s)/(1 + s) \ge 1/3$, this proves (7). Retaining the entire tail is what avoids a loss of a factor s in this estimate.

Combining the two bounds shows that

$$J_{r,q}(s) \le \frac{3^q (2\pi)^{qs}}{\pi} s^r \int_0^{\pi} [L_0 + \log_+(1/k)]^r k^{-qs} \, \mathrm{d}k.$$

On $(0, 1)$, the substitution $t = -\log k$ bounds the last integral by $\int_0^{\infty} (L_0 + t)^r e^{-t/2} \, \mathrm{d}t$ when $qs \le 1/2$. On $[1, \pi]$ it is at most πL_0^r. Also $(2\pi)^{qs} \le (2\pi)^{1/2}$. These bounds provide an explicit finite choice of $K_{r,q}$. For $q = 0$ the same calculation omits the gap factor. □

We shall need the following immediate consequences:

$$S_2 = D^{*2}(0) = \int_{-\pi}^{\pi} d(k)^2 \frac{\mathrm{d}k}{2\pi} = \frac{\zeta(2 + 2s)}{2Z^2} = \frac{\pi^2}{12} s^2 + O(s^3), \tag{10}$$

$$I_{2,1} := \int_{-\pi}^{\pi} \frac{d(k)^2}{1 - d(k)} \frac{\mathrm{d}k}{2\pi} = S_2 + O(s^3), \tag{11}$$

$$J_{2,3} = O(s^2), \qquad J_{3,2} = O(s^3). \tag{12}$$

For (10), (5) implies $Z^{-1} = s + O(s^2)$, and the convergent series $\sum_{n \ge 1} (\log n) n^{-2}$ gives $\zeta(2 + 2s) = \zeta(2) + O(s)$. For (11), use

$$I_{2,1} - S_2 = \int_{-\pi}^{\pi} \frac{d(k)^3}{1 - d(k)} \frac{\mathrm{d}k}{2\pi}, \qquad |I_{2,1} - S_2| \le 2J_{3,2},$$

since $1 - d(k) \le 2$. None of these estimates assumes $d(k) \ge 0$.

3 A nonbacktracking lower bound

For this section define the pointwise moments

$$S_m = \sum_{x \ne 0} D(x)^m = 2^{1-m} \frac{\zeta(m(1 + s))}{Z^m} \qquad (m \ge 2).$$

Only S_2 is identified here with a convolution return probability; in particular, $S_3 = \sum_x D(x)^3$ is not $D^{*3}(0)$.

Proposition 3.1. *For sufficiently small $s > 0$, the equation*

$$F_s(p) := \sum_{x \neq 0} \frac{pD(x)}{1 + pD(x)} = 1 \tag{13}$$

has a unique solution $p_{\mathrm{NB}} \in (1, 2)$. Moreover,

$$p_c \geq p_{\mathrm{NB}} = 1 + S_2 + O(S_3 + S_2^2) = 1 + S_2 + O(s^3). \tag{14}$$

Proof. Fix $p > 0$ with $F := F_s(p) < 1$, and put $q_x = pD(x)$ and $f(x) = (1 + q_x)^{-1}$ for $x \neq 0$. On bounded nonnegative functions of a previous displacement define the positive operator

$$(Bh)(x) = \sum_{y \neq 0,\, y \neq -x} q_y h(y).$$

It is bounded in the supremum norm by p. Symmetry gives

$$Bf(x) = F - \frac{q_x}{1 + q_x} = f(x) - (1 - F) \leq Ff(x),$$

where the last inequality uses $f(x) \leq 1$. Writing $q_{\max} = \sup_x q_x$, we have $1 \leq (1 + q_{\max})f$, so positivity and induction imply

$$B^{n-1}1 \leq (1 + q_{\max})F^{n-1}f.$$

The total product weight W_n of nonbacktracking walks of length n starting at zero consequently satisfies

$$W_n = \sum_x q_x (B^{n-1}1)(x) \leq (1 + q_{\max})F^n \qquad (n \geq 1). \tag{15}$$

All these sums are nonnegative and finite; for example $W_n \leq p^n$.

Every self-avoiding path is nonbacktracking, and its bond-open probability is the product of its bond probabilities because its unordered bonds are distinct. A union bound therefore bounds the probability of an open self-avoiding path of length n from zero by W_n. We do not assign product probabilities to the nonbacktracking walks that repeat bonds; their weights only enlarge the numerical upper bound. Each vertex has finite open degree almost surely, since its expected degree is $p < \infty$. By countability the random graph is locally finite almost surely. An infinite component then has self-avoiding paths of every finite length from each of its vertices. Equation (15) proves absence of an infinite cluster whenever $F_s(p) < 1$. The case $p = 0$ is immediate.

The function F_s is continuous and strictly increasing, by domination by $pD(x)$ on bounded intervals. The exact expansion

$$F_s(p) = p - p^2 S_2 + p^3 T_3(p), \qquad T_3(p) = \sum_x \frac{D(x)^3}{1 + pD(x)}, \quad 0 \leq T_3(p) \leq S_3, \tag{16}$$

shows $F_s(1) < 1$. If $a = S_2 \leq 1/8$, then

$$F_s(1 + 2a) \geq 1 + 2a - (1 + 2a)^2 a = 1 + a(1 - 4a - 4a^2) > 1.$$

There is thus exactly one root in $(1, 1 + 2a)$, and $p_c \geq p_{\mathrm{NB}}$. The interval lies below $2 < 2Z$, so all parameters used are admissible. For instance $0 < s \leq 1/4$ ensures $S_2 \leq \zeta(2)s^2/2 < 1/8$.

Set $x = p_{\mathrm{NB}} - 1 \in (0, 2a)$. Equation (16) yields

$$x = p_{\mathrm{NB}}^2 a - p_{\mathrm{NB}}^3 T_3(p_{\mathrm{NB}}), \qquad |x - a| \leq (4a + 4a^2)a + 8S_3.$$

Since $S_3 \leq \zeta(3)s^3/4$, the asserted uniform expansion follows. □

4 The infrared input and the required diagram estimates

For fixed s, write $\tau_p(x) = \mathbb{P}_p(0 \leftrightarrow x)$ and $\chi(p) = \sum_x \tau_p(x)$. We use the standard equality of the susceptibility threshold and the percolation threshold, in the form stated for this long-range model in [3, Section 1.2.5]. Consequently $\chi(p) < \infty$ for every $p < p_c$. This sharpness input is distinct from a definition of p_c by susceptibility.

For percolation, Assumption 2.1 of [3] requires a symmetric probability kernel with $D(0) = 0$, together with

$$\|D\|_\infty \le \beta, \qquad \int_{-\pi}^{\pi} \frac{d(k)^2}{[1-d(k)]^3} \frac{\mathrm{d}k}{2\pi} \le \beta, \tag{17}$$

for sufficiently small β. The kernel (1) has full support off zero, in particular it contains the nearest neighbors. Lemma 2.1 verifies (17) with $\beta = B_0 s$ for a fixed B_0 and all sufficiently small s. Here B_0 is independent of s; it is chosen before reducing the interval of s. Theorem 2.3, equation (1.35), and the threshold bound (2.6) of [3] then give constants K_0, K_1 such that

$$1 \le p_c \le 1 + K_0 s, \tag{18}$$

$$0 \le \widehat{\tau}_p(k) \le \frac{1 + K_1 s}{\chi(p)^{-1} + p[1 - d(k)]}, \qquad p < p_c. \tag{19}$$

The constants in this use of the theorem are fixed by its smallness threshold and the dimension $d = 1$, not by an additional spread parameter. The parameter $s = 3$ in its percolation assumption is a diagram exponent; it is unrelated to our variable $s = \sigma$.

For completeness, the Fourier nonnegativity in (19) is a property of percolation connections. For every finitely supported complex sequence (a_x),

$$\sum_{x,y} a_x \overline{a_y} \tau_p(y - x) = \mathbb{E}_p \sum_{\mathcal{C}} \left| \sum_{x \in \mathcal{C}} a_x \right|^2 \ge 0,$$

where the sum is over the open clusters. Thus τ_p is positive definite; since it is in ℓ^1 subcritically, its continuous Fourier transform is nonnegative. This does not assert nonnegativity of $d(k)$. For the rest of the proof we restrict to $1 \le p < p_c$, where

$$\widehat{\tau}_p(k) \le \frac{b_s}{p[1 - d(k)]}, \qquad b_s = 1 + K_1 s, \quad k \ne 0. \tag{20}$$

The single point $k = 0$ has no effect on the integrals used below.

Suppress p from τ_p and put $q = pD$. Define

$$T(x) = (q * \tau^{*3})(x), \quad T_* = \sup_x T(x), \quad T_0 = T(0), \quad A = \tau^{*3}(0), \quad B = \tau^{*2}(0). \tag{21}$$

All these quantities are finite at a fixed subcritical parameter.

Lemma 4.1. *Uniformly for $1 \le p < p_c$ and sufficiently small $s > 0$,*

$$T_* = O(s), \qquad T_0 = O(s^2), \qquad 0 \le A - 1 \le 3T_0, \qquad 0 \le A - B \le 3T_0, \qquad B = 1 + O(s^2). \tag{22}$$

Proof. Set $h = \tau - \delta \ge 0$. The first-edge connection bound gives $h \le q * \tau$: for a nonzero endpoint choose the first bond of a simple open path, remove that bond, and use independence and a union bound. Since $\tau \ge \delta$,

$$\tau^{*3} - \delta = h * (\tau^{*2} + \tau + \delta) \le 3h * \tau^{*2} \le 3q * \tau^{*3} = 3T. \tag{23}$$

Convolving once with q shows

$$T(x) \le q(x) + 3(q^{*2} * \tau^{*3})(x).$$

The Fourier integrand $p^2 d(k)^2 \widehat{\tau}(k)^3$ is nonnegative. Fourier inversion therefore bounds the last convolution at every x by its value at zero. By (20),

$$U := (q^{*2} * \tau^{*3})(0) \le \frac{b_s^3}{p} J_{2,3} = O(s^2). \tag{24}$$

Thus $T_* \le p\|D\|_\infty + 3U = O(s)$ by (18), while $T_0 \le 3U$ because $D(0) = 0$. Equation (23) at zero gives $A - 1 \le 3T_0$. Finally $1 \le B \le A$, so both remaining assertions follow. □

In particular, the closed value T_0 is smaller than the spatial supremum T_*. Keeping that distinction will turn the error terms below into $O(s^3)$, whereas a bound by T_*^2 alone would give only $O(s^2)$.

5 A uniform upper bound on the first lace coefficient

The decomposition below follows van der Hofstad and Sakai [4, Section 3.2, (3.45)–(3.55)]: nonzero first attachment, terminal membership in a deleted-bond cluster, and deletion of the first bond. The additional estimate is the separation $T_* = O(s)$ from $T_0, A - 1, A - B = O(s^2)$. It improves the remainder to $O(s^3)$ in this changing-exponent regime. The reusable bound to keep in view is (40): an exact subtraction $p^2 S_2$ and nonnegative errors each carrying an extra small factor.

We specify the source events before estimating them. This also records which restrictions survive the diagram bounds. Write $x \Leftrightarrow y$ if $x = y$ or there are two bond-disjoint open paths from x to y. For a set $\mathcal{A}$, the event $\{x \overset{\mathcal{A}}{\longleftrightarrow} y\}$ means that x and y are connected and every open path between them has a bond with an endpoint in $\mathcal{A}$, or that $x = y \in \mathcal{A}$. This is the "through" convention of [2, Definition 3.1]. Let $\widetilde{\mathcal{C}}^{\{u,v\}}(x)$ be the cluster of x when bond $\{u, v\}$ is forced vacant.

A directed bond (a, b) is pivotal for $v \leftrightarrow x$ if its removal disconnects v from x, with a on the v side and b on the x side. Let $\mathcal{P}(v, x)$ be the set of these directed pivotal bonds. Define

$$E'(v, x; \mathcal{A}) = \{v \overset{\mathcal{A}}{\longleftrightarrow} x\} \cap \left\{\nexists (a, b) \in \mathcal{P}(v, x) : \{v \overset{\mathcal{A}}{\longleftrightarrow} a\}\right\}. \tag{25}$$

On the connection event in this definition the pivotal bonds are open. The event E' itself need not be increasing. We use the van den Berg–Kesten (BK) inequality only on the increasing connection events in its cited diagrammatic upper bounds.

Use subscripts $0, 1, 2$ for independent percolation configurations at the same parameter. With the set from level 0 held fixed in the level 1 probability, the first coefficient is

$$\pi_p^{(1)}(x) = \sum_{u,v} q(v - u)\, \mathbb{E}_0 \left[\mathbf{1}_{\{0 \Leftrightarrow u\}} \mathbb{P}_1 \left(E'(v, x; \widetilde{\mathcal{C}}_0^{\{u,v\}}(0))\right)\right]. \tag{26}$$

This is [2, (3.31)], also used in [4, Section 3.2, (3.44)–(3.45)], with each bond weight replaced by its probability $q(v - u)$. Set $\overline{\pi}_p^{(1)} = \sum_x \pi_p^{(1)}(x)$.

Proposition 5.1. *There is a constant K_2 such that, uniformly for $1 \le p < p_c$ and $0 < s < \sigma_0$,*

$$0 \le \overline{\pi}_p^{(1)} \le (2p - p^2) S_2 + K_2 s^3 \le S_2 + K_2 s^3. \tag{27}$$

5.1 The two restricted error terms

For $u = 0$ abbreviate $\mathcal{C}_v = \widetilde{\mathcal{C}}_0^{\{0,v\}}(0)$ and $\tau_v^{\mathrm{del}}(x) = \mathbb{P}_p(x \in \mathcal{C}_v)$. Split the sum in (26) into the three disjoint cases

$$u \ne 0; \qquad u = 0,\ x \notin \mathcal{C}_v; \qquad u = 0,\ x \in \mathcal{C}_v. \tag{28}$$

The last part is at most

$$M_p = \sum_{v,x} q(v)\tau_v^{\mathrm{del}}(x)\tau(x-v), \tag{29}$$

by dropping the no-pivotal requirement, retaining only the connection $v \leftrightarrow x$ at level 1, and using independence of the two levels.

To bound the first two parts, let $\circ$ denote disjoint occurrence with disjoint open-bond witnesses. The source events [2, (4.2), (4.6)] for $N = 1$ are

$$F_0(0,u,w,z) = \{0 \leftrightarrow u\} \circ \{0 \leftrightarrow w\} \circ \{w \leftrightarrow u\} \circ \{w \leftrightarrow z\},$$
$$F_1(v,t,z,x) = \{v \leftrightarrow t\} \circ \{t \leftrightarrow z\} \circ \{t \leftrightarrow x\} \circ \{z \leftrightarrow x\}.$$

Their terminal inclusion is

$$E'(v,x;\mathcal{C}) \subseteq \bigcup_{z\in\mathcal{C}} \bigcup_{t} F_1(v,t,z,x). \tag{30}$$

In the joint event of (26), the initial double connection and the cluster membership in this inclusion are bounded by F_0 at level 0, as in [2, (4.10)]. In particular the vertex u in $F_0(0,u,w,z)$ is the original u of (26); it has not been summed out or renamed. In words, the terminal inclusion retains a vertex in the specified cluster. If the endpoint lies outside that cluster, these two vertices cannot coincide; this strict exclusion must survive the later upper bounds.

Put $\widetilde{\tau} = q * \tau$ and

$$A_3(a,b,c) = \tau(c-a)\tau(b-a)\tau(c-b).$$

The union bound, BK for F_0, F_1, and summation over v give for the $u \neq 0$ part

$$\sum_{u\neq 0,w} A_3(0,u,w) \sum_{z,t,x} \widetilde{\tau}(t-u)\tau(z-w)A_3(z,t,x). \tag{31}$$

For each u, w, Cauchy–Schwarz gives $(\tau * \tau)(r) \le B$, and therefore

$$\sum_{z,t,x} \widetilde{\tau}(t-u)\tau(z-w)A_3(z,t,x) = \sum_{z,t} \widetilde{\tau}(t-u)\tau(z-w)\tau(t-z)(\tau*\tau)(t-z)$$
$$\le BT(w-u) \le BT_*.$$

Also $\sum_{u,w} A_3(0,u,w) = A$. The restriction $u \neq 0$ excludes the entire $u = 0$ slice, whose sum is $\sum_w \tau(w)^2 = B$. Consequently

$$\text{the first part of (28)} \le BT_*(A-B) = O(s^3). \tag{32}$$

For the second part, (30) supplies a vertex $z \in \mathcal{C}_v$. The condition $x \notin \mathcal{C}_v$ therefore forces the precise restriction $z \neq x$. Retain it before dropping $x \notin \mathcal{C}_v$. Since $\mathbb{P}_p(z \in \mathcal{C}_v) \le \tau(z)$, BK bounds this part by

$$\sum_{v,z,t,x:\, z\neq x} q(v)\tau(z)\tau(t-v)\tau(t-z)\tau(t-x)\tau(z-x). \tag{33}$$

For $r = t - z$, the strict terminal sum is

$$H(r) = \tau(r)\big[(\tau*\tau)(r) - \tau(r)\big] \ge 0, \qquad \sum_r H(r) = A - B. \tag{34}$$

Indeed, the removed $x = z$ term has value $\tau(r)^2$. Summing v, z in (33) now gives

$$\sum_r H(r)(q * \tau^{*2})(r) \le T_*(A-B) = O(s^3), \tag{35}$$

where $q * \tau^{*2} \le q * \tau^{*3}$ follows from $\tau \ge \delta$. Together with (29), this proves the uniform reduction

$$\overline{\pi}_p^{(1)} \le M_p + O(s^3). \tag{36}$$

All error constants in this reduction follow from Lemma 4.1; no estimate on a critical coefficient is used.

5.2 The deleted bond and the main convolution

Let $F_p = (q * \tau^{*2})(0)$. Deletion reduces connection probabilities, so $M_p \le F_p$. A necessary improvement comes just from the terms $x = v$. Conditioning on bond $\{0, v\}$ gives the exact identity

$$\tau(v) = q(v) + [1 - q(v)]\tau_v^{\mathrm{del}}(v) \qquad (v \neq 0). \tag{37}$$

The parameter range in use has $\|q\|_\infty = O(s) < 1$. Retaining the improvement at $x = v$, where $\tau(x - v) = 1$, yields

$$\begin{aligned} M_p &\le F_p - \sum_v q(v)[\tau(v) - \tau_v^{\mathrm{del}}(v)] \\ &= F_p - p^2 S_2 + R_p, \qquad R_p = \sum_v q(v)^2 \tau_v^{\mathrm{del}}(v). \end{aligned} \tag{38}$$

The nonnegative remainder is cubic:

$$0 \le R_p \le \|q\|_\infty \sum_v q(v)\tau(v) = \|q\|_\infty (q * \tau)(0) \le \|q\|_\infty T_0 = O(s^3). \tag{39}$$

This is an exact deleted-bond subtraction; expanding an inverse vacancy probability is unnecessary.

Before making asymptotic estimates, the preceding event bounds give the single explicit inequality

$$\boxed{\overline{\pi}_p^{(1)} \le F_p - p^2 S_2 + \|q\|_\infty T_0 + (B + 1)T_*(A - B).} \tag{40}$$

The two terms after the subtraction are nonnegative and each is $O(s^3)$ by Lemma 4.1. Both strict restrictions leave the factor $A - B$. More sharply, the positive first-edge inequality gives $A - B = (h * \tau^{*2})(0) \le T_0$ and $B - 1 \le 2(q * \tau^{*2})(0) \le 2T_0$.

To estimate F_p, use $h = \tau - \delta \le q * \tau$ and $D(0) = 0$:

$$\begin{aligned} F_p &= 2p(D * h)(0) + p(D * h * h)(0) \\ &\le 2p^2 (D^{*2} * \tau)(0) + p^3 (D^{*3} * \tau^{*2})(0). \end{aligned} \tag{41}$$

The first Fourier integrand below is nonnegative. Applying (20) with its factor p retained gives

$$p^2 (D^{*2} * \tau)(0) \le p b_s I_{2,1}, \tag{42}$$

$$0 \le p^3 (D^{*3} * \tau^{*2})(0) \le p b_s^2 J_{3,2}. \tag{43}$$

For (43) the convolution is nonnegative in real space, but its Fourier integrand $p^3 d(k)^3 \hat{\tau}(k)^2$ can have either sign. Taking its absolute value before using (20) is what justifies the bound.

Equations (11), (12), and (18) now imply

$$F_p \le 2p b_s I_{2,1} + p b_s^2 J_{3,2} = 2p S_2 + O(s^3).$$

Combining this with (36)–(39) proves

$$\overline{\pi}_p^{(1)} \le (2p - p^2) S_2 + O(s^3).$$

Finally $2p - p^2 = 1 - (p - 1)^2 \le 1$. This proves Proposition 5.1, including its uniformity over the full subcritical interval $[1, p_c)$.

6 The positive finite-order remainder and the threshold

We use the finite-order lace identity, rather than an infinite signed series. Let

$$\pi_p^{(0)}(x) = \mathbb{P}_p(0 \Leftrightarrow x) - \delta(x), \qquad \Pi_1 = \pi_p^{(0)} - \pi_p^{(1)}.$$

The identity at truncation order $M = 1$ is

$$\tau = \delta + q * \tau + \Pi_1 * q * \tau + \Pi_1 + \mathcal{R}_1, \qquad \mathcal{R}_1(x) \geq 0. \tag{44}$$

This is [2, (3.9)–(3.10), (3.32)]; the weighted infinite-lattice version is [3, Section 4.2, (4.6)]. For clarity, its positive remainder has the form

$$\mathcal{R}_1(x) = \sum_{u,v,a,b} q(v-u)q(b-a)\,\mathbb{E}_0\Big[\mathbf{1}_{\{0\Leftrightarrow u\}}\mathbb{E}_1\Big[\mathbf{1}_{E'(v,a;\widetilde{\mathcal{C}}_0^{\{u,v\}}(0))} \times \mathbb{P}_2\big(\{b \xleftrightarrow{\widetilde{\mathcal{C}}_1^{\{a,b\}}(v)} x\}\big)\Big]\Big]. \tag{45}$$

Dropping the through-cluster constraint at level 2 gives the pointwise bound from [2, (3.36)]:

$$0 \leq \mathcal{R}_1 \leq \pi_p^{(1)} * q * \tau. \tag{46}$$

The cited construction applies to subcritical infinite graphs, where clusters are almost surely finite, and its probability identities allow inhomogeneous independent bond weights. We use HHS's weighted infinite-volume identity directly. Connection-event BK bounds on the countable bond set can equivalently be obtained by finite witness approximation. No monotone convergence assertion for the nonmonotone event E' is needed. For each fixed $p < p_c$, all sums in (44) are absolutely convergent: $\pi_p^{(0)} \geq 0$ and

$$\sum_x \pi_p^{(0)}(x) \leq \chi(p) - 1 < \infty,$$

Proposition 5.1 bounds the sum of $\pi_p^{(1)}$, and (46) gives $\sum_x \mathcal{R}_1(x) \leq p\overline{\pi}_p^{(1)}\chi(p) < \infty$. The signed convolution is absolutely summable by these same bounds.

Set

$$U_s = S_2 + K_2 s^3, \qquad a_p = 1 + \sum_x \pi_p^{(0)}(x) - \overline{\pi}_p^{(1)}.$$

Reduce σ_0 so that $U_s < 1/2$. Nonnegativity of $\pi_p^{(0)}$ and Proposition 5.1 imply $a_p \geq 1 - U_s > 0$. Summing (44) gives the exact scalar identity

$$\chi(p)[1 - pa_p] = a_p + \sum_x \mathcal{R}_1(x) > 0. \tag{47}$$

Since $\chi(p)$ is positive and finite, for every $1 \leq p < p_c$,

$$p < \frac{1}{a_p} \leq \frac{1}{1 - U_s}. \tag{48}$$

If $p_c = 1$ the same upper bound is automatic; otherwise take the supremum over the scalar parameters $p \in [1, p_c)$. This yields

$$p_c \leq \frac{1}{1 - S_2 - K_2 s^3} = 1 + S_2 + O(s^3), \tag{49}$$

because $S_2 = O(s^2)$ and hence $U_s^2 = O(s^4)$. There is no passage to a critical Fourier transform, susceptibility divergence, or a critical limit of a lace coefficient in this argument. Avoidance of coefficient continuity is not a new general goal: Liu [5, Appendix A] gives a different argument at

criticality through a convolution identity and nonsummability. Here the positive finite truncation gives a scalar inequality at every subcritical parameter. The saving is in the additional correction calculation, not a replacement for the lace-expansion work underpinning the imported infrared theorem.

The lower bound in Proposition 3.1 and (49) prove (2). Formula (10) then proves (3). To make the uniform quantifiers explicit, choose σ_0 smaller than $1/6$, the fixed smallness threshold needed for (17), and the fixed thresholds used for $\|q\|_\infty < 1$ and $U_s < 1/2$. Every estimate above then holds for every real $0 < s < \sigma_0$ and every $p \in [1, p_c(s))$, with common constants. This completes the proof of Theorem 1.1.

7 A two-scale criterion and additional kernel families

The proof separates two scales rather than requiring a specific zeta kernel. The following consequence makes that distinction reusable. All kernels in this section define independent undirected bonds with the exact linear law $pD(x)$.

Theorem 7.1 (Two-scale kernel criterion)**.** *Let D_ϵ be even probability kernels on $\mathbb{Z}$, with $D_\epsilon(0) = 0$, positive nearest-neighbor weights, and $d_\epsilon(k) = \widehat{D}_\epsilon(k) < 1$ for $0 < |k| \le \pi$. Suppose the admissible percolating set is nonempty and, with common constants as $\epsilon \downarrow 0$,*

$$\|D_\epsilon\|_\infty = O(\epsilon), \qquad \int_{-\pi}^{\pi} \frac{|d_\epsilon|^2}{(1-d_\epsilon)^3} \frac{\mathrm{d}k}{2\pi} = O(\epsilon^2), \qquad \int_{-\pi}^{\pi} \frac{|d_\epsilon|^3}{(1-d_\epsilon)^2} \frac{\mathrm{d}k}{2\pi} = O(\epsilon^3).$$

Then

$$p_c(D_\epsilon) = 1 + \sum_x D_\epsilon(x)^2 + O(\epsilon^3).$$

The implicit constant is uniform over families with common hypothesis constants and a common sufficiently small parameter interval.

Proof. Write $a = J_{2,3}$, $b = J_{3,2}$ and $m = \|D_\epsilon\|_\infty$. Since $0 \le 1 - d \le 2$, $S_2 = \int d^2 \frac{\mathrm{d}k}{2\pi} \le 8a = O(\epsilon^2)$, and the pointwise third moment is at most $mS_2 = O(\epsilon^3)$. The proof of Proposition 3.1 uses only evenness, summability and these moment bounds. It supplies the lower bound with the stated error. Its test parameters are admissible because $m \to 0$.

Assumption 2.1 of [3] holds with $\beta = O(\epsilon)$. The same theorem used in Section 4 gives $p_c = 1 + O(\epsilon)$ and (20) with $b_\epsilon = 1 + O(\epsilon)$. Its dimension remains fixed at one. The proof of Lemma 4.1 consequently gives $T_* = O(\epsilon)$, $T_0 = O(\epsilon^2)$ and $A - 1, A - B, B - 1 = O(\epsilon^2)$. Moreover,

$$|I_{2,1} - S_2| \le 2J_{3,2} = O(\epsilon^3).$$

The event bounds in Section 5 use neither the power-law formula nor positivity of its Fourier transform. With these new bounds their exact deleted-bond inequality gives $\overline{\pi}_p^{(1)} \le (2p - p^2)S_2 + O(\epsilon^3)$ uniformly subcritically. Section 6 then gives $p_c \le (1 - S_2 - O(\epsilon^3))^{-1} = 1 + S_2 + O(\epsilon^3)$. All imported results, and the inherited event decomposition, are the same as for the original theorem; no new lace expansion is claimed. □

Theorem 7.2 (Summable positive additions)**.** *Fix $B_0 < \infty$. For any sequence $b_n \ge 0$ with $B = \sum_{n \ge 1} b_n \le B_0$, define*

$$D_s(\pm n) = \frac{n^{-1-s} + b_n}{2(Z_s + B)}, \qquad n \ge 1, \qquad D_s(0) = 0.$$

Uniformly over these sequences, for sufficiently small $s > 0$,

$$p_c = 1 + \frac{\zeta(2+2s) + 2\sum_{n\ge1} b_n n^{-1-s} + \sum_{n\ge1} b_n^2}{2(Z_s+B)^2} + O_{B_0}(s^3)$$
$$= 1 + \frac{s^2}{2}\left(\zeta(2) + 2\sum_{n\ge1}\frac{b_n}{n} + \sum_{n\ge1} b_n^2\right) + O_{B_0}(s^3).$$

No moment assumption beyond summability is imposed on the additions.

Proof. Let d_s^0 denote the original power-law characteristic function. The new kernel is a mixture with original-kernel weight $w_s = Z_s/(Z_s+B)$. Thus

$$|d_s(k)| \le |d_s^0(k)| + B/Z_s \le s[L_0 + B_0 + \log_+(1/|k|)], \qquad 1 - d_s(k) \ge w_s(1 - d_s^0(k)).$$

For small s uniformly in $B \le B_0$, $w_s \ge 1/2$, so the latter is at least $\frac{1}{6}(|k|/(2\pi))^s$. The integral proof of Lemma 2.1 applies with these modified constants, giving $J_{r,q} = O_{B_0}(s^r)$ for fixed r, q and $qs \le 1/2$. Also $\|D_s\|_\infty \le (1+B_0)s/2$. Full support off zero gives the structural hypotheses. To check that admissible percolation exists, at $p = 3$ the model dominates the original kernel at parameter $3w_s > 2$ for small s; Theorem 1.1 places that model above its threshold. The parameter 3 is admissible since $3\|D_s\|_\infty < 1$ eventually. Theorem 7.1 proves the first formula.

For the second, $\sum b_n^2 \le B_0^2$ and

$$\left|\sum b_n n^{-1-s} - \sum b_n/n\right| \le s\sum b_n \frac{\log n}{n} = O(B_0 s).$$

Together with $\zeta(2+2s) = \zeta(2) + O(s)$ and $(Z_s+B)^{-2} = s^2 + O_{B_0}(s^3)$, this proves the assertion. □

For example, $b_1 = \lambda \ge 0$ and $b_n = 0$ for $n > 1$ changes the coefficient from $\pi^2/12$ to $\pi^2/12 + \lambda + \lambda^2/2$. Thus the coefficient is sensitive to local connection weights even though the long-distance exponent is unchanged.

Theorem 7.3 (Compact mixtures of exponents)**.** *Fix $0 < c_- \le c_+ < \infty$. For any probability measure ν on $[c_-, c_+]$, let $D_s = \int D_{cs}^0\, \nu(\mathrm{d}c)$, where D_t^0 is the power-law kernel (1). Uniformly over ν,*

$$p_c(D_s) = 1 + \frac{\pi^2}{12}\left(\int c\,\nu(\mathrm{d}c)\right)^2 s^2 + O_{c_-,c_+}(s^3).$$

Proof. For $c_+ s \le 1/2$, integration of Lemma 2.1 gives

$$|d_s(k)| \le c_+ s[L_0 + \log_+(1/|k|)], \qquad 1 - d_s(k) \ge \tfrac{1}{3}(|k|/(2\pi))^{c_+ s}.$$

Thus $J_{r,q} = O(s^r)$ when $qc_+ s \le 1/2$, and $\|D_s\|_\infty \le c_+ s/2$. Full support again gives the structural conditions. For admissible percolation, integral comparison yields

$$\frac{D_{cs}^0(n)}{D_{c_+ s}^0(n)} = \frac{Z_{c_+ s}}{Z_{cs}} n^{(c_+ - c)s} \ge \frac{c_-}{c_+(1+c_+ s)} \ge \frac{c_-}{2c_+}$$

for small s. The fixed parameter $p = 6c_+/c_-$ is eventually admissible and dominates an original-kernel model at parameter at least 3. Apply Theorem 1.1, then Theorem 7.1. Finally Tonelli's theorem gives

$$S_2 = \iint \frac{\zeta(2+(c+d)s)}{2Z_{cs}Z_{ds}}\,\nu(\mathrm{d}c)\nu(\mathrm{d}d).$$

Uniformly on the compact parameter square, $Z_{cs}^{-1} = cs + O(s^2)$ and $\zeta(2+(c+d)s) = \zeta(2) + O(s)$. This proves the formula. □

8 Scope and further questions

The correction in Theorem 1.1 is the two-step return weight of the exact normalized kernel. The proof requires both the uniform small-exponent Fourier bounds and the percolation event restrictions; evaluating random-walk return probabilities alone would not identify the lace coefficient. The positive odd-truncation remainder is enough for an upper bound on the threshold, so a matching asymptotic formula for $\overline{\pi}_p^{(1)}$ and a bound on every higher coefficient are not part of the argument.

The result fixes dimension one and lets only σ tend to zero. The error constant is not evaluated numerically, and the proof gives neither a third-order coefficient nor a useful quantitative estimate throughout $0 < \sigma < 1$. Other bond laws, including exponential connection probabilities, may differ at the correction scale and are not covered by the theorem. The argument also does not determine the oriented critical correction. The uniform kernel estimates and the one-sided lace estimate yield the sufficient criterion and the two additional families in Section 7. Signed additions, unbounded exponent ratios and useful numerical error constants remain outside those statements.

Declarations

Data and reproducibility. This is an analytic study. No numerical experiment or empirical data is used in the proof. The model, estimates, and imported results are specified in the text; the manuscript source and bibliography form the associated research materials.

Ethics. The work involves no human participants, animals, personal data, or experimental intervention.

Contributions and AI disclosure. OpenAI Codex was used to develop the arguments, compare the cited sources, perform internal mathematical checks, and prepare this draft. The scholarly attribution is Anonymous. Evidence Press is the publisher; account ownership is not a claim of human mathematical contribution. The supplied review and current checks are AI-assisted scrutiny, not external human peer review or formal verification.

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Archive boundary. The original submission contains internal file-hash bindings but omits four historically cached external-source files and its original packaging test suite. Those historical bytes and executions cannot be reconstructed from that ZIP. Current source checking and any new diagnostics are recorded separately; neither is a replacement for the missing historical records.

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